Conditional Probability & Two-Way Tables

Section 1: Two-Way Frequency Tables

1 A survey of 100 students asks whether they prefer online or in-person learning: 60 are High Schoolers (40 online, 20 in-person) and 40 are Middle Schoolers (15 online, 25 in-person). What fraction of all students surveyed prefer online learning?

a. $\frac{11}{20}$

b. $\frac{9}{20}$

c. $\frac{2}{5}$

d. $\frac{3}{5}$

2 In the same survey, what is the probability that a student chosen at random is a Middle School student who prefers in-person learning?

a. $\frac{1}{4}$

b. $\frac{2}{5}$

c. $\frac{3}{10}$

d. $\frac{1}{5}$

3 A hospital clinic logs 120 patients: 70 adults (30 tested positive) and 50 children (20 tested positive). What is the probability that a randomly chosen patient tested negative?

a. $\frac{5}{12}$

b. $\frac{7}{12}$

c. $\frac{1}{2}$

d. $\frac{2}{3}$

Section 2: Conditional Probability from Frequency Tables

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4 From the student survey (60 High School: 40 online, 20 in-person; 40 Middle School: 15 online, 25 in-person), given that a randomly chosen student prefers online learning, what is the probability that the student is in High School?

a. $\frac{8}{11}$

b. $\frac{2}{3}$

c. $\frac{3}{11}$

d. $\frac{4}{5}$

5 Given that a student is in Middle School, what is the probability that they prefer in-person learning?

a. $\frac{5}{8}$

b. $\frac{3}{8}$

c. $\frac{1}{4}$

d. $\frac{1}{2}$

6 At a conference, 80 attendees are engineers (50 use Python) and 40 are managers (10 use Python). Given that an attendee uses Python, what is the probability that they are an engineer?

a. $\frac{5}{6}$

b. $\frac{2}{3}$

c. $\frac{3}{4}$

d. $\frac{1}{6}$

Section 3: Independent and Dependent Events

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7 Events A and B are independent. If $P(A) = 0.40$ and $P(B) = 0.60$, what is $P(A \text{ and } B)$?

a. 0.24

b. 0.48

c. 0.52

d. 1.00

8 A jar contains 6 red marbles and 4 blue marbles. Two marbles are drawn at random without replacement. What is the probability that both marbles are red?

a. $\frac{1}{3}$

b. $\frac{9}{25}$

c. $\frac{2}{5}$

d. $\frac{1}{2}$

9 If $P(A) = 0.5$, $P(B) = 0.7$, and $P(A \text{ and } B) = 0.35$, which of the following is true?

a. A and B are independent events.

b. A and B are mutually exclusive events.

c. $P(A \mid B) = 0.7$

d. $P(B \mid A) = 0.5$

Section 4: Union & Addition Rule

10 A standard six-sided die is rolled. What is the probability of rolling an even number OR a number greater than 4?

a. $\frac{1}{2}$

b. $\frac{2}{3}$

c. $\frac{5}{6}$

d. $\frac{1}{3}$

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11 If $P(A) = 0.65$, $P(B) = 0.45$, and $P(A \text{ and } B) = 0.25$, what is $P(A \text{ or } B)$?

a. 0.80

b. 0.85

c. 0.90

d. 1.10

12 In a group of 50 students, 30 play soccer, 25 play basketball, and 10 play both sports. How many students play neither sport?

a. 5

b. 10

c. 15

d. 20

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